CFD interpretation practical: transonic airfoil screening

Instructions and reference material

You have **30 minutes**. Work individually. No calculator, network access, AI tool, code, or notes are permitted. Show a short calculation where requested; concise evidence-based statements are sufficient elsewhere.

Scenario

A 625 kg high-subsonic cargo UAV, including its payload, is screened at one flight condition. The results below come from a two-dimensional inviscid finite-volume airfoil calculation. Three 12%-thick candidate configurations have different camber and incidence.

Reference relations

Use $g = 10$ N/kg and the supplied load factor $n = 1.2$:

$$W = mg, \qquad L_{\text{req}} = nW, \qquad L = q_\infty S C_L, \qquad D = q_\infty S C_D.$$

For this problem, $q_\infty S = 10.0$ kN. The strong-shock screening limit is

$$M_{\text{surface,max}} \leq 1.60.$$

A rapid decrease in Mach number paired with a sharp pressure recovery in a surface C_p curve indicates a shock. A strong shock can raise transonic wave drag and may cause separation in a real viscous flow.
Model boundary. The CFD coefficients are used here as a simplified wing-screening approximation. They do not certify viscous drag, separation, three-dimensional effects, structural loads, stability, or full aircraft performance.

1. Requirements and preliminary recommendations [10 points]

Candidate	Camber	α	C_L	C_D	$M_{\text{surface,max}}$
A: NACA 4412	high	2.25°	0.958	0.166	1.70
B: NACA 2412	moderate	1.75°	0.799	0.110	1.56
C: NACA 1412	low	1.50°	0.627	0.079	1.46

Engineer A: "Select A. It has the greatest lift coefficient and therefore the greatest payload margin."
Engineer B: "Select C. It has the smallest drag coefficient and therefore is the most efficient choice."

1. (2 points) Calculate L_{req} and $C_{L,\text{req}}$.

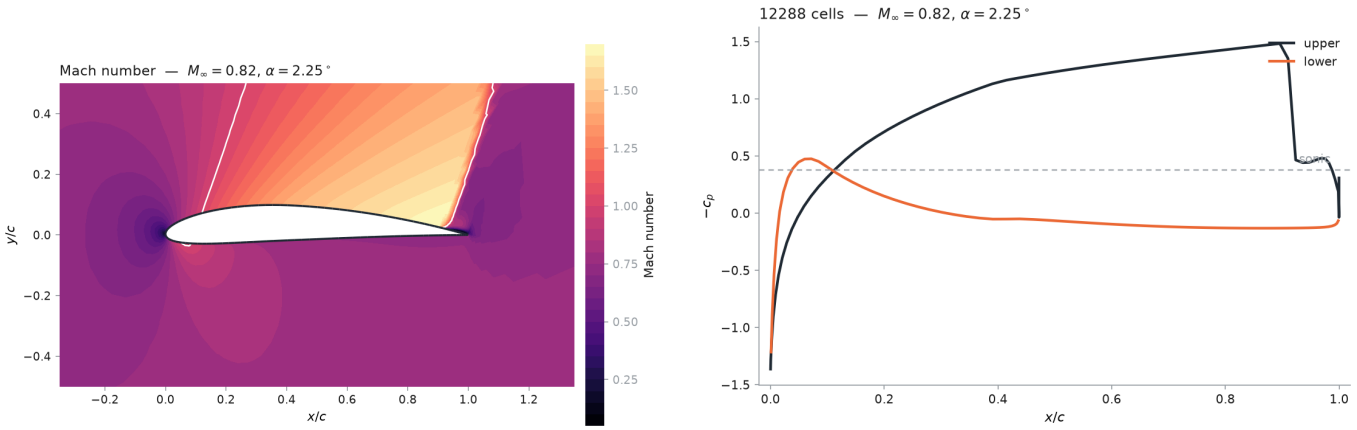
2. (2 points) Calculate Candidate B's predicted lift and drag.

3. (4 points) Complete the compliance matrix.

Candidate	Meets lift requirement?	Meets strong-shock limit?	Keep/reject
A			
B			
C			

4. (2 points) In one sentence each, state why Engineer A's and Engineer B's recommendations are incomplete.

2. Flow-field evidence [7 points]



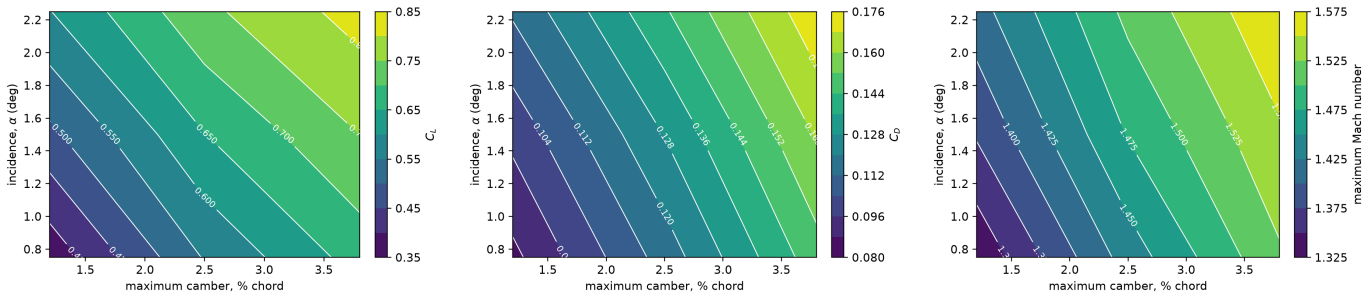
Candidate A: NACA 4412 at $M_\infty = 0.82$ and $\alpha = 2.25°$. These are the custom course results used in this practical.

5. (2 points) On the supplied Mach and C_p plots, mark the shock location.

6. (3 points) Explain how increasing aerodynamic loading can increase lift while also producing a stronger shock and higher drag. Cite one observation from a Mach contour and one observation from a C_p curve.

7. (2 points) Explain why Candidate A must be rejected despite its large C_L.

3. JAX sweep and design recommendation [9 points]



Custom coarse JAX-adjoint study: fixed 12% thickness and camber crest at $x/c = 0.40$. The red contour marks $C_L = 0.30$ or $M_{\max} = 1.30$.

8. (2 points) Identify a feasible region of the (m, α) sweep: it must meet both the lift and strong-shock requirements.

9. (2 points) The supplied custom results use one mesh. State the next numerical check required before treating the drag ranking as reliable, and explain why.

10. (5 points) Complete the design record below.

Recommendation: I recommend Candidate _____ / a nearby point in the feasible region because _____.

It meets the payload-weight requirement because _____.

The next simulation or experiment should be _____ because _____.